ActInf Livestream #007.1: "Variational ecology and the physics of sentient

Livestream | 54:06

all right
hello and welcome everyone to the active
inference
live stream this is active inference
live stream 7.1
and it is october 27th 2020.
welcome to team com everyone we are an
experiment in
online team communication learning and
practice
related to active inference you can find
us on
twitter at inferenceactive at our gmail
you can find us on key base or youtube
as well
this is a recorded and an archived live
stream so please provide us feedback so
that we can improve our work
all backgrounds and perspectives are
welcome here and as far as video
etiquette for live streams
mute raise your hand so that we can get
to everyone in the queue
use respectful speech and try not to use
the jitsi chat
if possible today in act imp
7.1 we're going to start with a warm-up
just briefly introducing ourself and
checking in
and then we'll work through some of the
sections of 7.1
the paper we're going to be discussing
is variational ecology and the physics
of sentient systems by ramstead at all
in 2019

and today we'll talk about the goals
abstract the roadmap
potentially move through a few key
quotations and then
preview the figures mainly with a focus
on seeing
what each figure represents and today in
7.1 and next week in 7.2 where we may or
may not be in costume
we will discuss this paper so please
save and submit your questions whether
you're on the panel now or whether
you're watching in replay
and get in touch if you'd like to
participate
so for the intros and check-ins it'd be
great if everyone could just
introduce themself and their location
say hello in a short intro and then pass
it to somebody who
hasn't spoken so i'm daniel i'm in
california
and i'll pass it to a first time
discussion cameron
hi yeah i'm cameron um
i'm here in zurich in switzerland i just
started a
phd in philosophy of mind at the
university of zurich
first time at this group um
shannon
hi everyone i'm shannon i'm a phd
student at the university of california
in merced
um i'll pass it to lee
hi everyone uh i'm in london a phd
student at the university of
york um a bit of an active inference

beginner
and i will pass it to alex
hello uh i'm in moscow russia
i'm a researcher at systems management
school
yeah and the opposite session
hello um uh my name is sasha i'm
in california and i'm a neuroscience
graduate student
um and i will pass it to
muddy hi i'm muddy
um i'm uh the director of ai at uh
versus labs in the spatial web
foundation uh
my link to active inference is uh i i
did my phd with karl
riston four or five years ago still very
much a beginner
and i'm currently in uh rainy liverpool
and so i will pass it to
anybody who would like to volunteer
because i can't see the screen i
apologize
um hello it's stephen here i'm in
toronto i'm doing a practice-based phd
at canterbury christchurch university in
the uk
and i'll pass it over to
maxwell hey everyone i'm maxwell
ramstead
i'm a postdoctoral fellow at mcgill
university in montreal
i'm skyping to you from the montreal
suburbs
um and yeah i work on uh multi-scale
extensions of active inference i'm the
um first author on
on this paper that we'll be discussing

uh very grateful
to have such a great group of people as
discussants
and i will pass it to i believe lee is
the only one who hasn't spoken yet right
no no i've been cool yeah i think that
gets
through it perfect sasha has no yeah
she has okay sorry yep great thanks
everyone for the intros
so in these two warm-up questions any
thoughts related or unrelated
anything that kind of gets you excited
and helps on-ramp to the discussion
we'll be having
what is a physical or informational
ecosystem
that you care about or have studied and
while people
are raising their hand um the first
ecosystem that came to mind for me was
in southeast arizona where there's a
grassland ecosystem where there's a lot
of cool ants and also the grazing
agricultural use by cows had a big
effect on the ecosystem kind of pushing
it from a grassland regime
into a very um different sort of pebbly
regime
so when i was thinking about this
variational ecology i was kind of
thinking about
the desert ecosystem and and the
southwest
and go ahead muddy
um so i'm interested in a slightly
different non-biological
ecosystems uh but still ecosystems where

we have
agents that make decisions so we build
our micro service meshes by
having multiple agents that can make
decisions and can decide for themselves
whether they
are appropriate to a given computation
or not so we treat them as ecosystems
and we
are using active inference very much to
model the the life cycles of these
various micro services
cool and if anyone else wants to raise
their hand muddy i'm actually
just kind of curious let's go to steven
first and then i'll follow up if
possible yeah i'm interested in kind of
intentional spaces but learning spaces
or workspaces and one
thing that really came up for me was
when i was in a place called
inguavuma at poncini school
back in 2004 when i visited there
and to do a project or just at that time
just to engage the community
i it was at a school then ponchini
school in a rural area
where they'd only just built the school
10 years before so they didn't actually
have schools there before that
and um in in and
you know around the outside you've got
the fence with the barbed wire on the
top
in the middle of this area with no
electricity and it was
and i i i sensed at that time there was
this kind of disconnect between

the students understanding what this
niche was that they were
immersed in and an area where
um you know while they're learning stuff
there's also 35
hiv and big challenges
and and this kind of learning
environment which was kind of
coming from a colonial background so i
this this kind of
difference between the world which i was
seeing in that community in the world in
this
this school kind of you know as i've
seen in certain workplaces you see this
kind of disconnect this kind of like
learning isn't just the knowledge it's
something about how to fit it together
and that's when i so later became this
idea of a social topography whether
somehow what's that kind of landscape of
meaning and
when is it more than just knowing the
facts and i think that's where the
ecosystem
sort of comes together cool i like how
we're exploring a lot of these types of
ecosystems and those are definitely what
we want to
encompass in the variational ecology
discussion
sasha
um yeah so i'm interested in the micro
environment
that neurons grow up in and
how that affects their activity levels
and how they make connections with
others

so one of the things that i'm studying
for that
is how different immune molecules
and turning on immune genes
affects this process and this is
something that i want to learn from this
paper um how to apply that to my own
system
but one parallel i'm also thinking about
is
uh the immune landscape um
when people are isolated and aren't
exposed to as many people what their
kind of immunity is like um
especially in a time when we're really
thinking about our own health and
immunity
ooh just from writing down some of these
little notes there's ecosystems across
scale
that's definitely going to come back and
then there's this idea of community
ecology
which in the bio sense is usually about
different species
but then there's of course as steven was
talking about this group and historical
dynamics to our community ecology
and then ecosystems that are of
different technologies
so the second warm-up question is what
is something that you're curious about
or want to explore in today's discussion
and while people are raising their hand
i think the question that almost
motivated or seems to motivate this
paper is the one i'm curious about which
is about applying

the free energy principles formalisms to
larger scale systems where potentially
the stability of the markov blanket or
some of the other dynamics of the system
are so rapidly changing that it um
really questions how we're going to
study it from this framework
and i know that's something that we'll
return to
muddy yeah i i think what's fascinating
um from from this work that maxwell has
been
doing is is helping us actually get
really fine-grained detail onto the
computational currents of certain
problems such as
you know when do i stop modeling my
subsystem and when do i start
representing my subsystem as
a parameter so when you know this
transition between a parametric approach
and a modeling approach to then feed
into a hierarchical model
that is uh quite a it's quite an art and
it's an
ill-posed sort of problem so i think
this work is really really fascinating
from that perspective
from me personally cool maybe
maxwell i'd be curious what is that
relationship with the
parametric and the modeling approaches
like how do we go about
exploring that or what does that mean
from a modeler's perspective
i'm not sure i understand your question
um like
muddy was talking about how this uh

question of modeling like when to stop
and
when to include a parameter at one level
versus another what's
an introductory thought or what's the
rationale or the motivation behind doing
that kind of a thing
i mean so mel andrews has thought about
this a lot more than i have
i mean uh i wish they were here
um that they're sick today so i mean
roughly speaking the generative models
that we employ under active inference
uh they harness the what mel has been
calling the existential variables of the
system
uh so you know the the variables that
actually make a difference
to uh survival uh
so you know you can think of stuff like
a core body temperature
and so on i mean ultimately there is
always going to be an arbitrariness
uh involved in the modeling effort
because we have to make
kind of science science side decisions
about what are the relevant things that
we're going to model or not
but that said there do seem to be some
some variables that are more critical to
the continued existence
of the systems that we're interested in
so i don't know if that addresses the
point
cool i think we'll return to it
especially as we think about
what are the vital signatures or the
vital variables for ecosystems because

we want to be thinking about them in
this
similar framework cool here we go
so the paper that we're going to be
discussing in 7.1 today
and in 7.2 last week as well as in 7.0
last week
uh yeah people can figure that one out
is
variational ecology in the physics of
sentient systems by ramstead
constant badcock and firsten and the
goal of the paper
was stated as reviewing a framework for
modeling complex adaptive systems for
multi-scale
free energy bounding organism niche
dynamics
thereby integrating the modeling
strategies and heuristics
of variational neuroethology or vne with
a broader perspective on the ecological
nestedness of biotic systems
we extend variational neuroethology
beyond the action perception loops of
individual organisms i.e active
inference
by appealing to the variational approach
to niche construction
to explain the dynamics of coupled
systems constituted by organisms
and their ecological niche and so i
quoted that directly from the paper
because i think it's a really
direct representation and foreshadows
the fusion
of the vne and the va nc which we can
talk about

but to kind of rephrase that in non-fep
language it's a question about how can
we make
a multi-level or multi-scale ecology
framework
that builds upon eco evo devo that's
ecology evolution and development
but also adding in insights from physics
mathematics and complexity or complex
adaptive systems
so it certainly is an ambitious scope
and it's kind of linking together this
organism and
downward story of the variational
neuroethology
with the niche construction aspect so
kind of we have the organism looking
downwards that's the neuromolecular
components of behavior
and then we have the organism looking
upwards and outwards with the ecology
yeah if i could comment on that briefly
yeah so the basically the if you're
looking at like the history of how this
came about
um so i was working
i mean between 2000 i guess 16 and 18 i
was mostly focused
on developing uh you know this
multi-scale
active inference story in large part
with carl fristen
and that resulted in a framework that we
called a variational ethology or
variational neuroethology
in our 2018 physics of life reviews
paper
um answering short ingress question free

energy formulation
and so the the variational neuro
mythology is essentially
uh a theory about the kind of if you if
you want to think about it like
um visually a kind of vertical stack
of systems we've discussed this on this
podcast before but like this idea that
systems are made of nested systems of
systems so that
essentially if you look at the ontology
of a system what you have is a stack of
segregated
systems of different spatial spatial and
temporal scales
that affect effectively compose each
other so cells composing
tissues composing organs composing
organisms composing
social groups and composing species and
so on
and so following the publication of that
paper i started working closely with
axel constant
who uh was riffing off work that yella
brinaberg had been doing
in amsterdam uh with his group over
there
and uh axel and i developed um
a basically a niche construction account
of
active inference what we highlighted was
the fact that active inference is a
symmetric formulation in the sense that
it tells a story about how internal and
external states attune to each other
internal states of tuning to external
ones through things like perception and

and learning and phenotypic
accommodation and development
and external states tuning to internal
states through action notably
making the world more like uh the way
that we expect it to be
so that gave us a kind of horizontal
dimension where basically
at every kind of level of this uh
vertical stack
which you have is effectively a
relationship between
uh an ecological niche and the denizens
or inhabitants
of that ecological nation so
this paper uh aimed to combine both
frameworks uh so the niche construction
stuff and the nested systems of system
stuff
into a kind of principled uh approach to
biological systems
uh and so you know moving from an
ethology
and a theory of niche construction so
ethology being the study of animal
behavior
essentially so combining these two
things into a bona fide
theory of ecology uh using the
variational free energy principle
i'm sure you're going to mention this
after but the last ingredient that we
added to this mix
is the skilled intentionality framework
from the amsterdam group
eric groupveld julian kieverstein
uh yellow greenenberg what they add to
the mix

with the skilled intentionality stuff is
connecting this predictive
processing framework to the the
ecological psychology
notion of affordances um yes so this
paper essentially takes
the ecological psychology stuff on
affordances the so-called skilled
intentionality framework
the variational ethology and the
variational niche construction stuff and
proposes this integrative model
uh yeah so that's the kind of history
awesome
great summary thanks so much maxwell and
just to kind of
restate that in this nested framework
you have the
bottom up or emergence features of the
system you have the top-down
features of the system and then the
lateral are kind of like the collective
behavior
so we're thinking about all these
ingredients that maxwell just mentioned
especially coming from the skilled
intention intentionality framework and a
few other areas bringing them all
together
so let's check out the abstract this
paper addresses the challenges faced by
multi-scale formulations of the
variational
or free energy approach to dynamics that
obtain for large scale ensembles such as
ecosystems or
groups of organisms we review a
framework for modeling

complex adaptive control systems for
multi-scale free energy
bounding organism niche dynamics thereby
integrating the modeling strategies and
heuristics of variational
neuroethology which remember was really
focused on the organism's feedback with
its local affordances
with a broader perspective on the
ecological nestedness of biotic systems
so like a lot of times in this podcast
we've talked about the affordance of
getting a jacket or of changing the
thermostat but that's actually
part of the constructed niche of humans
it's not just a human in a box it
doesn't find themselves without
affordance
unlike these some of the biomechanical
affordances perhaps
we extend the multi-scale variational
formulation beyond the action perception
loops of individual organisms
by appealing to the variational approach
to niche construction
to explain the dynamics of coupled
systems constituted by organisms and
their ecological niche
and that relates to the symmetry that
was being discussed a few minutes ago
we suggest that the statistical
robustness of living systems is
inherited
in part from their eco-niches as niches
help coordinate dynamical patterns
across
larger spatiotemporal scales we call
this approach variational ecology

so it's kind of like the affordance of
driving in a car
is part of the constructed niche of the
highway network and then that
shapes the movement patterns we argue
that
when applied to cultural animals such as
humans variational ecology enables
us to formulate not just the physics of
individual minds but also a physics of
interacting minds across spatial and
temporal scales
of physics of sentient systems that
range from cells to society
and this should sort of invoke that
warm-up discussion about the different
kinds of ecosystems
especially when talking about humans and
our technologically enabled extended
niches
part of the ecosystem is informational
it's symbolic
and so we also want to be thinking about
information ecologies in the way that
these dynamics apply not just to the
ants on the pebbles
but also to humans and the paths through
the parks that are discussed in the
paper
and also ways that long-range
technological communication
might come into play so just to briefly
run through the road map this paper is
zero indexed which i found very
interesting and it starts with a
discussion
of the variational or free energy
formulation

and just briefly reviews active
inference and generative models it's
really great that every paper
sort of starts with this uh section on
just
what it is that we're talking about with
active and fvp
they're so annoying to write though
honestly you know you don't want to you
don't want to plagiarize you don't want
to just copy the paragraphs you've
written
uh it's like how many ways can you
explain the same thing you know
and it's like a if there really is a
smooth
energy landscape underlying these
explanations
we're treading many many paths and
exploring many on-ramps turning these
ideas around through discussion and
through
more papers so yeah there's a repetition
element and there's also
potentially a way that we make bigger
and more accessible explanations by
rephrasing it
again and again for different journals
or different specialties
so then the second section is on
variational neuro anthology and
that brings into question uh multi-scale
levels of analysis as well as ensembles
of markov blankets as well as a few
figures that we'll scan through
the third section introduces the sort of
organism
and upwards perspective that is given by

the niche construction the variational
approach to niche construction
and relates that to ontologies of
affordances some of which we've talked
about in previous discussions as well
another figure on the bayesian mechanics
and active inference
and then as usual with these papers
there's sort of the
foundation then idea a and idea b
and then i a and b are linked together
and so we get to
section four variational ecology the
physics of shared mites
and this also relates to discussions
we've had about
what it is that's modeling is the
organism a model of its environment does
have a model and agreed that mal's
thoughts on this are
super helpful and then there's some
concluding remarks
and just to really quickly visually
outline this
theory by addition approach that's being
enacted here
on the very bottom we have the fep which
is that paradigmatic ground rock
that a lot of these other theories are
going to be built on top of
then from left to right we have active
inference tin bergens four
questions which are explored a little
bit in
number 7.0 as well as some previous
discussions we've had
est is evolutionary systems theory which
is basically

evolutionary biology with a systems
perspective and eco evo devo
which is also a very interesting and
rich area that emphasizes that
the way that evolution occurs is not
just ecologically embedded and enacted
but it occurs through changes in
development so you don't just get a
longer femur
you get the developmental proclivity to
have your growth plate be open for a
longer or shorter time
resulting in a phenotype that is
changing but it's changing through
development
active inference and tin brokens for
questions and evolutionary systems
theory
were basically summated this is not a
perfect summation it's just sort of
building it up a little bit to vne which
was a very organism and biobehavioral
centric approach to explaining action
perception loops of organisms
on the other hand also drawings from
actinf was the
variational approach to niche
construction which drew a little bit
more extensively from the eco
evo devo motivating questions and
literature
and then to bring in synergetics which
is this relationship between the higher
order
uh parameters the bigger slower things
and then the lower level parameters the
faster and smaller things
of hakan 1983 on the left side uh

and then the collective behavioral
approach these lateral relationships
between nested systems
it all kind of comes together to get to
variational ecology
which hopefully builds upon the
strengths of all of these different
antecedent theories
and gives us something that is going to
allow us to dip back
into for example behavioral explanations
of organisms or
dip into the niche construction approach
or the collective behavior across
different levels of analysis as well as
providing a helpful
path back to understanding how previous
formulations
for example a modern synthesis type
evolutionary biological explanation
or a non-fep eco evo devo explanation
it gives us an interpretation path that
we can walk
to understand a lot of the previous
literature so let's just
pause here does anyone have any thoughts
on this
theory summation or is there any other
areas that we think might be relevant to
draw in
one that i was wondering about is this
introduction of the minds
and the sentient systems and so there
was a sentence in the paper about the
sentient systems being
sensory systems though often when people
talk about the search for sentient life
and things like that

they're talking about the experiential
dynamics as well and so
the idea of getting the interacting
minds
from the interacting evolutionary
systems
i just thought that was kind of a
curious thing maybe foreshadowing in
integration with psychology or with
phenomenology
well so the the sentience aspect is
really i think just to highlight what
the
the free energy principle stuff does i
mean at the end of the day
uh the free energy principle shares with
perceptual control theory the idea that
action is about
uh the control of perceptual input
namely keeping it within certain bounds
um so that might be relevant to uh
to consider i mean yeah so from that
point of view sentience
is uh the capacity to be affected
uh by the world in a way that
allows us to control how the way
the world affects us yep like something
hitting a rock still could be felt or
is is still as influential on the state
of the rock but
it doesn't necessarily have the ability
to then act especially in a proactive
way
to get around that and also just one
other note is
actually the schrodinger's question
paper
from 1944 and then the answering

schrodinger's question paper from 2018 i
think are also
great places for people who are kind of
coming at this from a first principles
of biology perspective because
schrodinger's question was about how
organisms
are organized and very interestingly and
nine years before the discovery of the
structure of dna
and many years before a lot of the other
approaches that would come on the scene
with a molecular revolution
in the second half of the 1900s
schrodinger said
there has to be something like a
quasi-periodic crystal
and organisms have to organize at least
locally
they're not going to violate the laws of
physics this isn't vitalism but at least
locally you have to be able to take you
know the
bag of sugar and then organize it into
biomolecules that constitute an organism
that's a
increase in order how does that happen
is that the defining principle of life
and in some ways as we look on earth and
elsewhere for life these two
ideas of schrodinger really uh
remain as prescient as ever so i find
that just really interesting
stephen i know you're probably going to
mention this but i suppose the idea of
ergodicity sort of comes in through the
active inference
peace more than it normally thought of

in the other areas
but that maybe synergetics isn't
another way that these kind of ergodic
um ways to extract information at
present and i suppose the question is
is that non-ergodic processes the ways
to tap into things which are non-ergodic
and sort of causal
is that something which is when you then
step
outside of normal active inference and
you have this extra
structure on top which is our cognition
which allows us to
sort of process things that are kind of
non-ergodic so i think that might be
interesting like in terms of how we
construct things as humans as opposed to
that kind of more organic niche
construction but that
not that that's really touched on here
but i suppose it's just just the idea of
how patterns are extracted and what
levels of meaning or what processes of
like attuning or constructing or
relating what what's that kind of
dynamic at play
thanks for that comment i think the word
ergotic is really what a lot of that
idea hinges on so if anyone wants to
have a
take at what the word ergotic means i
tried to go through it a little bit
in 7.0 but um
if anyone wants to explain what an
ergotic
system is and stephen what your question
gets at is really whether the models of

ergotic systems
of systems where the time average is
equivalent to the ensemble or the
spatial averaging
um though there's also other ways to
phrase that or perspectives on that
in systems with extreme historiosity
like niche construction if you build a
road in one place the whole society kind
of leverages around that
or if there's an introduction of a new
species into an area
that can irreversibly and extremely
historically change the trajectory that
ecosystem
so how are we to model these systems
using ergotic frameworks
steven and then anyone else who wants to
step in on the ergotic
question because it's kind of a key term
yeah i was i think this is a really
interesting i thought of a relevant
point
because the thing with ergodicity is i i
sort of was
looking at the video for this session
that you made and sort of comparing it
to what i've been thinking is
it could be that there's lots of small
patches or periods or blocks where
there's kind of
enough ergodicity due to mixing
to extract information or to find
patterns
so it could be that there's not this
idea that the system is ergodic
but there might be lots of times where
there's moments of mixing that

ergodicity is present and you can
extract
or make inferences from that so i think
that's quite
um that changes things a little bit in
terms of how
you know makes it more plausible maybe
for how ergodicity
can creep in um
and it also i think then starts to give
you a rationale for how you
can engage with the whole entropy
question as well and
because normally entropy is brought into
this equation
talking about multiple levels and things
dissipating and suddenly it's like well
um if you've got periods of ergodicity
then you can you you've got a way
to extract or
get get a grip on the noise that's going
on out there over time
um and and i think that that that's
important i think that's also important
in terms of
trying to explain it to other people
because most people
when they think about the world and
systems i'm actually
there's a whole debate on the vibrant
systems
group about this is that they kind of
they just take a snapshot of perspective
on a system
and they say this is what the system
does but
that's not the same as something which
is happening over time

and there's a pattern over time it's
like it's a different way of thinking
about
the world and even think about systems
in general so
that that that piece is really important
about how does it happen over time and
is it lots of
bits of ergodicity that add together or
is
is it like we have to have ergodicity
over 100 years over 10 years over five
years
nice let's go shannon and then muddy and
then maxwell if you want
i i would just like to jump in quickly
just to clarify a few things about
ergodicity if that was okay
uh let's go yeah you might actually
answer my questions before i ask them
perfect shannon muddy go ahead okay so
ergodicity from a mathematical point of
view
you can understand it heuristically as
the
you know so a system is ergotic if
the average of our measures of the
system
converges to a measure of the average of
our the value of whatever we're
measuring so
another way of thinking about it is that
like wherever your system starts in its
state space
with an ergotic system it will end up in
in a regime of characteristic states
like so
so no matter where you start you end up

uh
in in these uh in these states so i mean
you know you can think of
uh you know my the my
my body temperature for example would be
describable in this
way you know it varies a bit but it ends
up
continuously converging to 36 and a half
degrees celsius
so there's a lot of confusion in the
literature and in discussion
about this ergodicity thing um one thing
to point out
is that uh so to the extent that it is
implied in the framework uh the
the notion that we're considering is
only local ergodicity so we're not
saying that
uh you know because the maximal version
of like an ergotic system
like a system that's globally ergodic is
a system that doesn't really have a
history
in the sense that it doesn't really
matter where you start you always end up
in the same place
um the notion of ergodicity that we're
employing in the free energy framework
to the extent that it's employed at all
is the notion of local eradicity which
means that
uh on some appropriate temporal
and spatial scale uh it's useful to
think about the system
as being ergotic as in that it will
regularly converge to the same
uh set of states or values of those

states rather
um so i guess there's a lot to say here
on on the one hand um you know this
local ergodicity thing
uh is important so when we say that the
earth is
uh is round you know
uh we're we're not saying that it's not
flat enough locally for us to build
buildings on it
uh so like locally the the earth has
it has fl is flat enough has a
negligible enough curvature that we can
build for example a building on it
and this is the same kind of idea so the
the ergodicity at play
is that you know on some appreciable
time scale
systems converge to the same set of
values effectively
and uh yeah so i guess that's
uh one point another related point is
that
ergodicity actually isn't um
they well it's it's not really baked
into the free energy principle
all you really need uh to get the free
energy principle running is a markov
blanket
and a non-equilibrium steady state or
generative model which we've been
discussing over the last few weeks
so generally notions like ergodicity and
weekly mixing
also appears in some of these
discussions it only applies to certain
kinds of systems with
multiple multiple particles under

special assumptions
namely the assumption that all the
particles are exchangeable
um so if we're dealing with a single
particle this kind of probability
distribution in general doesn't exist
the only probability distribution
uh to which we have access is when we
sample the state of a single particle
over time
um so yeah the the real
assumptions on which the free energy
stuff is built
is this non-equilibrium steady-state
assumption um
i mean and it's related to the
ergodicity thing in this in a certain
sense
um so non-equilibrium steady state we're
we're saying that you know the system
has a set of characteristic states to
which it it
tends you know over time uh
and so if if a system has uh
a non-equilibrium a steady state then
you know under the conditions that we
just described or
there are particles that are
exchangeable then uh you can treat the
system as ergotic
but it's it's not it's not as crucial to
the framework as some would suggest so
like i i've heard
in the you know in the bushes that uh
some people are mounting a critique of
the free energy principle based on the
idea that
uh living systems aren't actually

ergotic
i think that's a bit of a waste of time
because we're not we're not claiming
that
biological systems are globally ergodic
just that it's useful
to treat them as their gothic uh
at certain scales and you know the way
that we get get out of this kind of
weird
place formally speaking is by pointing
to this
nested systems of systems thing so the
idea is that
from the perspective of any layer of the
system the layer above it might as well
be stationary
so from the point of view of my bo of my
cells for example
my body is essentially stationary you
know so
their entire life cycle you know from
birth to uh cell division
uh will occur without any major
structural change happening to
the non-equilibrium steady state of my
whole body
so you know you you you get around
the implausibility of the strong
interpretation of these assumptions by
pointing out that well what's actually
at play is a nested
series of systems and you know
you you only need things to look ergotic
locally relative to other layers of the
system
i think that's really helpful for what
my question would have been

which would be like how would if you're
talking about the behavior of a crowd
and if there's a certain markov blanket
that can extend over this
this group of individuals um the thing
about a crowd is it's not a cell that's
not bounded by
being stuck inside of a body they could
leave like they could leave the stadium
or leave the protest
but if we're talking about ergotic or
like interacting just enough at just the
right spatial temporal scale then
we can talk about some markup blanket
over some
whole crowd or subset of a crowd at some
event at some time
exactly yeah that's precisely what's
going on here so
it's all about defining the relevant
spatial and temporal scales i mean if
your relevant spatial
and temporal scale is like you know the
whole city over several weeks then
obviously the the protesters that gather
you know and spontaneously form a social
group wouldn't count as
you know even having a non-equilibrium
steady state i mean it just kind of
forms and dissipates
but if you're looking over the scale of
you know minutes to
maybe hours then it does make sense to
say well locally
you know there's something robust here
it has a statistical independence from
its environment it seems to be
you know attracted to the same set of

states and then it dissipates
so so it's the same story over and over
again just a different kind of spatial
and temporal scale so you're totally
right i think shannon
that's right right on point cool let's
do muddy
and then stephen uh yeah i just wanted
to speak to some of the like just to
give a little bit of intuition so i'm
kind of
not super in active research right now
so these terms i i like
sharing kind of the way i think about
them for people who might be listening
who
might not be familiar with the like
ergodicity in general
so just generally i i and please correct
me if i'm wrong about this because i'm
looking to have my understanding probe
but i view ergodicity or a system that's
got it
um my intuition is that if i take a
sample of a system
that system is ergodic if that sample is
representative of the greater dynamics
of that system
so that's what maxwell was saying about
tending to the mean so it doesn't really
matter if i take my data from here here
here or here
as long as the sample i'm taking is
representative
um that there's another camera worms it
should tend to the same mean
or you know basically represent the same
dynamics of the underlying

model and underneath but the way in
which you define
ergodicity is actually defined by your
your measurement
your sample space it's necessarily
defined by your sample space so
you have all of these emergence
properties which are hard to pin down so
you have
like dynamic ergodicity which which are
you know partials with respect to time
which
tend towards some stable value for the
local and then the actual mean
versus static ergodicity which is if i
take a snapshot of my
system at any one given time so like a
cloud of gas
it kind of looks the same but without
any knowledge of my attractor systems i
can't say that my system is ergodic
dynamically over time
um and then you can think about
continual ergodicity at different levels
so um i think the example maxwell gave
with flat earth is a really really
beautiful one because if you take a
local representative sample say
100 meters by 100 meters
you you could define every single sample
by a single function
uh a stable function which either has
either a stability maximum point
or is linear in its nature so at that
level i have a continuous function which
defines
ergodicity across all scales but it's
part it's nested within a larger system

which might have
chaotic dynamics actually and so it's
it's actually ergodicity i think the
ambiguity comes in
when you're actually trying to define
the problem space itself
as a concept it's just basically saying
this sample represents the model i'm
trying this sample
of a larger system represents the
largest system
at whatever mathematical perspective i
happen to be looking at it from
whether it's with respect to time or
some other variable so i just wanted to
add that
in i mean just to play devil's advocate
thank you buddy for that i agree with
everything you said uh just to make
the uh you know put some meat on the
bones of criticism and the interest of
fairness
one of the reasons that you might resist
the idea that biological systems is
ergotic is for example
uh the pretty plain fact that evolution
isn't going anywhere
right there doesn't seem to be some kind
of evolutionary
kind of attractor basin towards which uh
all of evolution you know
uh contra this kind of anthropocentric
uh you know religious view of you know
man's place in nature well humans aren't
all that special um and
we're uh we're just like one among
uh you know a series of other
uh animals um yeah so

that wait one quick before we go to
stephen also if your body temperature
hits a certain very low or very high
value it's not returning to a steady
state either
and so that's like there's a lot of
reasons to think at a certain scale of
analysis
that there's violations of this kind of
behavior stephen i mean
very clearly we all die right yeah we
all dissipate and so
at long enough time scales nothing is uh
hasn't
has like this non-equilibrium steady
state or is ergodic in that sense right
like we all
we all just dissipate you know in some
sense the
the the only real ultimate ergotic
you know end point is the heat death of
the universe right so
yeah it's uncanny valley because at a
short time scale you're stable then at
the lifelong time scale you dissipate
and then
again you're just a flash in the pan so
very cool
and highlights that it's important what
scale we're talking about steven before
we close out the slide
yeah i also think that you know we talk
about the system is the system ergodic
internally externally but i don't know
whether this is right but i kind of
i'm more from a practice perspective
trying to do the sense making
of active inference and that kind of the

dynamics of
things going backwards and forwards so i
i'm trying to think of
the ergodicity as maybe more about the
process of this noisy
chaotic like muddy was saying data
going backwards and forwards or across
the markov blanket
and within there there's ergodicity
in that you know within and then that
even if the systems i don't know whether
they are or not ergodic
of the organism but there's something in
the data stream that allows you to
extract information from
the entropy the sort of the noise so i
was trying to work out if
you know that that might be two
questions here i think it ends up going
to this bigger question about whether
the systems are ergodic but i'm not sure
whether it's about
the dynamics the non-linear dynamics
have ergodicity sort of
tucked away in there yeah just to give
an ecological example let's think about
a predator prey relationship
like a latka volterra model you know the
cougars and the rabbits or something
like that
so it's always changing through time and
there's a winnerless competition where
they're always oscillating
and so from the first pass the first
moment it's not ergotic it's changing
at certain time scales but it has what
muddy was referring to as this dynamical
stationarity or the dynamical ergodicity

it's kind of like the more things change
the more they stay the same
sometimes when we're talking about these
higher order relational dynamics
or rates of change the partial
derivatives that muddy mentioned
then it's actually through time at the
correct time scale of analysis
that you see see that the predator prey
relationship is in like this
higher order ergodicity and then at
another time scale above that
the predator and the prey species don't
even exist so of course it can't be an
ergotic system so again
we see that there's time scales where
systems display ergodicity
and time skills and spatial scales where
they don't as well as relying on
questions about exchange ability
and other attributes but just in the
last
couple minutes this is really awesome
discussion
i want to just run through the figures
give people a little peek
into some of the formalisms and the
ideas that we'll be talking about
further in 7.2
and this is a quotation about
how exactly niche construction enters
the picture
and especially in terms of epistemic
resources which relates to the
technological niche of humans
as well as the skilled intentionality
framework
here's a good quotation about how in the

multi-scale framework
active inference is a group behavior
that's a group activity
and so this sort of connects the dots
between active inference and collective
behavior
because even when we draw the organismal
loop with active inference we're talking
about collectives of neurons or of
different
cell types and also ties it back to
closed causality circular causality
and synergetics this idea of
coordinating slower order parameters
and then rapidly oscillating higher
order parameters and that's kind of like
the
the dynamics of the gas cloud looks
stationary visibly but if you went down
there it'd be the seizing
seething mess and you'd say there's no
way this is a stationary state but it is
at a little bit of a higher level
and then um this last quote was just
about
how um at work one's behavior
acts to justify their sensory states and
their prior beliefs about them
and then that kind of reminded me of a
figure from the paper written with
alex and sasha and others where we
talked about that a little bit more
specifically from a
technological level um and
just to quickly walk through the figures
before we return
figure one marco blanket it's gotta be
figure one it's not it's not a ramstead

paper if it isn't
figure two is also unpacked a little bit
in the answering schrodinger's question
paper and this one there's what you see
and there's what you don't see
what you see is the potential
explanatory scope
across spatial and temporal scales that
range over many orders of magnitude
but what you don't see is that there's
nothing that's at the for example
one millisecond and one kilometer scale
maybe a lightning strike is able to
dissipate energy
rapidly at that scale but it doesn't
coordinate effectively it doesn't have
that closed
causality i should say uh but like
i no longer agree with this figure uh
casper has been i
starting may 2018 uh
i guess uh so casper uh was quick to
point out that you don't
you don't need a separate set of time
scales the
the spatial scales that we pointed out
are really
ontological scales so like the
subcellular scale
is a spatial scale and a temporal scale
uh simultaneously so
uh you know the cellular processes occur
over a few milliseconds and
yeah they have their specific size and
these these scales like
uh the the concomitant like
associated spatial and temporal scales
really increase along this diagonal

uh kind of together um so
i mean another thing that kind of breaks
this uh apart is where is culture here
right so culture is very fast uh but
it's also very large scale
a process so uh this is not a figure
that i would
recommend like learning by heart or
anything i think it turns out to be
wrong now but
science progresses right so yep we were
wrong
i still find it interesting especially
with that caveat that it's not like
there's small things happening
at the tiny scale and then there's no um
relationship with the niche construction
like niche construction for the cultural
niche and the communication niche
is related to cellular processes so
there's just what's happening
but then it's interesting to think about
how the functional levels of
organization
do align along this very
vague qualitative diagonal but not a
formal approach for sure here figure 3
is another recap figure having to do
with i think something about broccoli
and how it's a fractal now it's about
markov blankets and about nested systems
and how markov blankets at one level
can be thought of as engaging in
collective behavior at a higher level
figure four again an example that we've
seen before
with looking at how morphology can arise
as a function of initially equivalent

cells acting morphologically to reduce
their surprise about their expectations
about the extracellular gradients
and where they are and what that means
for what the organism is doing
and then how development and the
evolution of development could reflect
beliefs and action generative sequences
that
allow the organism to fulfill functional
morphologies
figure five i don't think is one that
we've seen yet
um but it connects the active inference
and a ford inside with a free energy
minimization
side and there's some mathematics there
that i hope to be actually going into
in the coming weeks as we've chosen the
paper for eight which is going to be
scaling active inference
and we'll be looking to have a lot of
discussion on the mathematics because
um it's definitely one of the areas we
want to dive deeper into
and then the last figure is a partition
of vector states
into particles which are the the um the
sets of balls here and um the blanket
states
so i'll go into that a lot next week
this is probably the key figure
perfect for the whole thing thanks for
noting that we'll definitely
go into this um in the uh last minute
thanks for participating in this short
and exciting 7.1
next time in 7.2 we're going to have

more time to talk about some of the
specific
questions that have been arising as well
as figure 6 as maxwell pointed out
everyone live or in replay is welcome to
send us comments or questions that would
be cool to unpack
live today i really appreciate just
hearing about how these different
ecosystem conceptualizations and
ways of thinking about the same
ecosystem same way of thinking across
different ecosystems how do they all
combine
and we provide follow-up forms to all
live participants so it's in the
calendar
invite and it'd be great if you could
fill it out we also
solicit feedback suggestions and
questions from
our audience and from our other
participants who aren't live here
and uh all we can say to that is
thanks for the participation thanks
everyone really
an awesome discussion and just as always
it's another
level of understanding another coat of
paint on the mural to hear from
everyone's perspective so i
deeply appreciate that and great times
anyone want to give a uh singular
closing thought to cap out the stream
long live ergodicity muddy go for it
oh i was just gonna say no
exactly can i ask you a question
that's my observation perfect thanks

everyone thanks again for listening for
participating and we will
see you next time